hidden units

X_1

$\vdots$

X_p

Activation function g

# of hidden units

# of layers

Called the "architecture"

W matrix of coeff for the first part

β matrix (vector) of coeff for the last part

Include more layers

Pick the # of units per layer

Each layer is activation function of linear combo of previous layer.

10 output variables for classification (later)

Two hidden layers

ex) L_1 has 256 units

L_2 has 128 units

$p = 785$

In this example

W_1 matrix of weights in 1st layer

$785 \times 256 \approx 201K$ weights to learn

Want to classify Y $\quad y \in \{1, \ldots, K\}$

label being predicted

Use the softmax activation $f_m(x)$

Values are non-neg.

Values sum to 1

Class predicted = one w/ $f_m(x)$ highest

$$f_m(x) = \frac{e^{Z_m}}{\sum_{\ell=0}^{9} e^{Z_\ell}},$$

Want to convert to probabilities

Same for any m

ex)

$e^1 + e^3 + e^{-1} + e^2 + e^5$

$= 178.97$

$$f_1(x) = \frac{e^1}{178.97}$$

$$f_2(x) = \frac{e^3}{178.97}$$

$$\vdots$$

$$f_5(x) = \frac{e^5}{178.97}$$

Output

$f = (0.015, 0.112, 0.002, 0.041, 0.829)$

Each data point : 28x28 grayscale image

Goal: predict the digit in the image.

entries
pixels
in [0,255]

Data converted into flat columns

Output predicts

$y_0, y_1, __, y_9$

60K training images 10K test images